Fiscal Year:	FY 2018	**Task Last Updated:**	FY 11/07/2018
PI Name:	Gerson, Stanton M.D.		
Project Title:	Contribution of GCR Exposure to Hematopoietic Stem Cell Dysfunction and Oncogenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2018 per S. Monk/HRP-LaRC (Ed., 12/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Radivoyevitch, Tomas Ph.D. (Case Western Reserve University) Welford, Scott Ph.D. (Case Western Reserve University)		
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Task Description:	Natural sources of radiation in space include galactic cosmic rays (GCR), solar energetic particles, and trapped energetic particles in a planetary magnetic field. These sources are difficult to shield because of their high energies and dense ionization patterns, thus posing significant health risks to astronauts on long term space missions. Efforts to protect astronauts from harmful cosmic radiation require a deeper understanding of the effects of GCR on human health. In particular little is known about the effects of GCR exposure on the hematopoietic stem cell (HSC) population and whether disrupted genetic stability in HSCs could result in the development of hematopoietic malignancies in astronauts on deep space missions. We propose to study the effects of high-LET (linear energy transfer) radiation characteristic of GCR on hematopoietic stem and progenitor cells in vivo and determine changes in HSC behavior that may predispose to leukemia and lymphoma development. Of importance is the fact that the average age of a shuttle crew has risen above 46 years, and our work and others have shown that HSCs display diminished function with age. Recent data from our group has demonstrated that middle-aged individuals show frequent defects in DNA mismatch repair in HSCs. We will therefore use a DNA repair deficient mouse model to study the effects of heavy ion radiation on HSCs that represent HSCs of potential astronaut populations, and determine whether exposure to high charge and energy (HZE) particles and the additional delta rays produced in their traversal through matter will disrupt the hematopoietic niche. We hypothesize that high-LET radiation characteristic of the GCR that will confront astronauts on space missions will damage HSCs and the HSC niche, disrupt HSC function, and contribute to induction and progression of hematopoietic malignancies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a cancer risk assessment study, the impact of our proposal extends beyond the benefit to NASA and its goal of risk quantification regarding the prospect of deep space travel to extraterrestrial bodies. Heavy ion exposure has a growing role in cancer therapy here on Earth through the use of hadron therapy for existing cancers. Our interests on the effects of heavy ions on repair compromised hematopoietic stem cells is thus directly relevant in the context of cancer treatment because irradiation of normal blood flowing through targeted tumor tissue could pose an additional risk to the development of second cancers involving circulating stem cells. The goals of our proposal will help quantify such risk both for astronauts and the general population.
Task Progress:	We demonstrated that Mlh1 deficiency does not affect short- and long-term HSC functions but lack of functional MMR (mismatch repair) does sensitize mice to lymphomagenesis in presence and absence of IR (infrared). The finding is very important to NASA due to the fact that future long-term missions will involve astronauts spending more time in the deep space. However, the tumorigenesis study was performed with only 56Fe ion IR and thus we decided to further strengthen our findings with two more HZE particles from GCR spectrum. Proton ion is the most prominent species of charged particle in GCR while 28Si ion is as frequent as 56Fe ion among HZE particles. Therefore, we performed HSC functional and tumorigenesis assays by irradiating Mlh1+/+ and Mlh1+/- mice with 10 or 100 cGy 1H ion (1000 MeV/n) or 28Si ion (300 MeV/n). To determine if Mlh1 deficiency affects HSC functions, we irradiated bone marrow (BM) cells collected from Mlh1+/+ and Mlh1+/- mice with 0, 10, 50, 100, or 250 cGy of 1H ion or 28Si ion. BM cells were plated to form myeloid colony forming units (CFU) and counted in order to determine impact of IR and Mlh1 status of HSCs. We discovered that high-LET 28Si ion caused more damage to BM cells and hence a significantly less myeloid CFU compared to 1H ion irradiated BM cells. As anticipated, Mlh1 status did not affect acute function of HSC measured by CFU assay. In addition, HSC differentiation was measured by complete blood count at 5 and 9 month post Mlh1+/+ and Mlh1+/- mice irradiated with 1H ion or 28Si ion. At these time points, irradiation impact on HSC differentiation was undetermined along with no effects of Mlh1 status on long-term differentiation. Next, we hypothesized that, due to Mlh1+/- mice showing a significantly higher and early incidence of lymphomagenesis in 56Fe ion irradiated mice, the impact of 1H ion or 28Si ion would be similar. Post IR, we followed mice up to 18 months and sacrificed any mouse with a visible sign of distress or palpable tumor. We found that Mlh1+/+ mice irradiated with any dose or IR source showed no significant increase in the tumor formation due to radiation exposure. In contrast, Mlh1+/- mice showed sensitization to radiation-induced tumorigenesis. In fact, we revealed a significant higher incidence of deaths due to tumor formation in Mlh1+/- mice irradiated with 100 cGy 28Si ion IR compared to 100 cGy 1H ion IR ($p=0.0001$). The finding shows that high-LET IR is more effective in prompting tumorigenesis compared to low-LET 1H ion IR in MMR compromised mice. In addition, histology and immunohistochemistry analysis revealed a majority of tumors in Mlh1+/- mice were lymphomas, in particular T-cell rich B-cell lymphomas. Collectively, the data revealed that incidence of lymphomagenesis in MMR deficient mice was LET dependent. To determine gene expression profile of each T-cell rich B-cell lymphoma cohorts, we decided to run RNA-sequencing analysis. The RNA-sequencing experiment was carried out to detect differentially expressed genes and to determine which signaling pathways are involved in the process of tumorigenesis. To study this, we collected RNAs from snap frozen T-cell–rich B-cell (TRB) lymphomas arising from Mlh1+/+ and Mlh1+/- mice irradiated with sham-, 1H ions, or 28Si ions. To identify differentially expressed genes, each TRB lymphoma gene expression profile was compared to Mlh1+/+ and Mlh1+/- lymph nodes collected from healthy age-matched sham-irradiated mice. The hierarchical clustering analysis revealed that all control age-matched lymph node samples clustered together with a similar gene expression profile. In contrast, TRB lymphomas gene expression profiles were discrete from control lymph nodes and no further distinction was observed between irradiation sources or Mlh1 status of TRB lymphomas. To visualized the similarity and differences between TRB lymphoma cohorts, we plotted the same gene expression profile data by grouping samples to their respective groups. The data showed that irradiated TRB lymphoma cohorts are distinct from sham-irradiated Mlh1+/- cohort, suggesting that tumorigenesis processes associated with irradiation are unique in nature. Further, we looked at the number of differentially expressed genes in each cohort to identify how many genes are differentially expressed between cohorts. We determined that over 85% of differentially expressed genes are found to be commonly associated with all cohorts, regardless of Mlh1 status or irradiation source. Interestingly, we observed that roughly 8% of differentially expressed genes are only associated with irradiated cohorts with no correlation found between low- and high-LET IR. Similarly, we also observed that roughly 11% of differentially expressed genes are only found to be associated with sham irradiated cohort. The data suggest that there are common genes involved in the process of tumorigenesis, but enough differences exist between irradiated vs sham-irradiated cohorts that could be the reason for early and higher frequency of tumorigenesis found in 28Si ion irradiated Mlh1+/- mice. In fact, gene ontology analysis revealed that processes, for instance DNA repair, mRNA and rRNA processing, ribosome biogenesis, and RNA

	splicing, were commonly involved in all TRB lymphoma cohorts. In contrast, differentially regulated genes belong to adherens junction and anchoring junction were predominantly found in irradiated TRB lymphoma cohorts, but not in sham-irradiated TRB lymphoma cohort. In conclusion, we showed that high-LET 28Si ion irradiation causes significantly early and higher incidence of lymphomagenesis compared to low-LET 1H ion irradiated Mlh1+/- mice. The RNA-sequencing analysis revealed that high number of differentially expressed genes are commonly found in each lymphoma. In addition, the analysis also detected that signaling pathways among the different cohorts were predominantly similar with a few signaling pathways distinct between the cohorts.
Bibliography Type:	Description: (Last Updated: 09/30/2020)
Articles in Peer-reviewed Journals	Patel R, Arakawa H, Radivoyevic T, Gerson SL, Welford SM. "Long-term deficits in behavior performances caused by low- and high-linear energy transfer radiation." Radiation Research. 2017 Dec;188(6):672-80. Epub 2017 Sep 29. https://doi.org/10.1667/RR14795.1 ; PubMed PMID: 28961076 , Dec-2017
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